

Recursive Functions

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- Due Monday by 11:59pm
- Points 10
- Submitting a file upload

Goal

This assignment will have you define and design two recursive functions. You will also review, at a very basic level, the use of a higher-order function that will apply each of these recursive functions to their arguments.

Instructions

Write the following two recursive functions:

```
def display_em(lower, upper):  
    """ This recursive function displays the consecutive integers  
    from its lower to its upper bounds """  
  
def add_em(lower, upper):  
    """ This recursive function calculates the sum of the consecutive  
    integers from its lower to its upper bounds """
```

These two functions will also serve as arguments to a higher-order function with the following signature:

```
def applyToEach(f, lower_bound, upper_bound):  
    """ This higher-order function applies the included function  
    to its lower and upper bound arguments """
```

The main program should, once again, prompt the user to enter the lower and upper bounds. It should also invoke the `applyToEach` function **twice** to output the results below by calling the appropriate recursive functions. The following includes a sample of the resulting output from running this program:

```
>>> %Run RecursiveFunctions.py  
Enter a lower bound: 10  
Enter an upper bound: 20  
  
The consecutive integers:  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20
```

```
Add up to 165
>>>
```

Save the source code week4Lab.py Take the all screenshot and save in to Microsoft word file name Wk4LabScreenshot.doc

Upload the above source code and test program source files and Wk4LabScreenshot.doc)

Grading

- This assignment is worth 10 points and will be graded using the attached grading rubric.

Helpful Resources

None.

- [Submitting your Canvas Assignment](https://community.canvaslms.com/t5/Student-Guide/How-do-I-submit-an-online-assignment/ta-p/503)  [\(https://community.canvaslms.com/t5/Student-Guide/How-do-I-submit-an-online-assignment/ta-p/503\)](https://community.canvaslms.com/t5/Student-Guide/How-do-I-submit-an-online-assignment/ta-p/503)

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Criteria	Ratings			Pts
This criterion is linked to a Learning OutcomeIntroductory documentation that includes the name and description of the program, and the name of the author.	1 pts Full Marks	0 pts No Marks		1 pts
Correct definition of display_em recursive function	3 pts Full Marks	2 pts Partial credit	0 pts No Marks	3 pts
Correct definition of add_em recursive function	3 pts Full Marks	2 pts Partial credit	0 pts No Marks	3 pts
applyToEach higher order function is correctly defined and used twice in the main program	2 pts Full Marks	1 pts Partial credit - it works	0 pts No Marks	2 pts
Program output has the same format as the example illustrated in the assignment	1 pts Full Marks	0 pts No Marks		1 pts
Total Points: 10				